

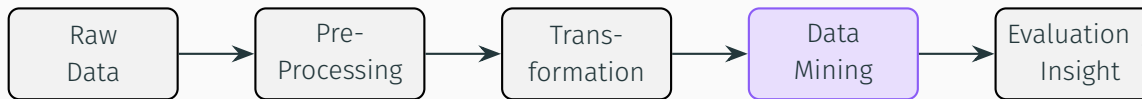
Data Mining & Unsupervised Learning

Discovering Hidden Patterns Without Labeled Data

Data Mining Concepts

The KDD Pipeline

Data Mining is a single, crucial step within the **Knowledge Discovery in Databases (KDD)** process.



- **Goal:** Extract non-trivial, implicitly valid, and actionable insights from large datasets.

Association Rule Mining

Market Basket Analysis seeks to discover combinations of items that frequently occur together.

Concept:

- We look for rules of the form:
 $X \rightarrow Y$
- "If a customer buys X, they are likely to buy Y."

Applications:

- Cross-selling
- Store layout optimization
- Recommendation engines

ID	Items
1	Milk, Bread, Diapers
2	Butter, Diapers, Beer
3	Milk, Diapers, Beer
4	Bread, Butter

Do Diapers $\rightarrow$ Beer?

Analyzing Rule Strength

To evaluate a rule $X \rightarrow Y$, we calculate three fundamental metrics:

1. Support

How frequently the itemset appears in the dataset. Filters out rare items.

$$\text{Support}(X \rightarrow Y) = P(X \cap Y)$$

2. Confidence

How often items in Y appear in transactions that contain X .

$$\text{Confidence}(X \rightarrow Y) = P(Y|X) = \frac{P(X \cap Y)}{P(X)}$$

3. Lift

The ratio of observed support to expected support if X and Y were independent.

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{P(Y)}$$

Unsupervised Learning

The Unsupervised Paradigm

Unlike Supervised Learning, Unsupervised Learning algorithms operate on datasets without target labels (y).

Primary Objectives:

- **Clustering:** Grouping similar data points together (e.g., Customer Segmentation).
- **Dimensionality Reduction:** Compressing the feature space while retaining variance (e.g., PCA).
- **Anomaly Detection:** Identifying outliers that do not conform to expected patterns (e.g., Fraud Detection).

The Challenge

Because there is no ground truth, evaluating the "correctness" of unsupervised learning models is inherently subjective and requires mathematical heuristics.

Clustering: The K-Means Algorithm

K-Means partitions N observations into K clusters by minimizing the Within-Cluster Sum of Squares (WCSS).

The Iterative Algorithm:

1. **Initialize:** Randomly place K centroids in the feature space.
2. **Assign:** Assign each data point to the closest centroid (using Euclidean distance).
3. **Update:** Move the centroids to the mean of the points assigned to them.
4. **Repeat:** Steps 2 and 3 until centroids stop moving (convergence).

Note: K-Means requires the practitioner to manually specify the number of clusters (K) before running.

Diagnostic: The Elbow Method

To determine the optimal K , we plot the WCSS against various values of K and look for the "Elbow".



Analysis: The elbow represents the point of diminishing returns. Adding more clusters beyond this point no longer significantly reduces the variance.

Dimensionality Reduction: PCA

Principal Component Analysis (PCA) mathematically transforms highly correlated features into a smaller set of uncorrelated features.

- **Principal Component 1 (PC1):** The line in the feature space that captures the *maximum* variance.
- **Principal Component 2 (PC2):** Captures the remaining variance, strictly orthogonal (perpendicular) to PC1.

Why do this? To combat the *Curse of Dimensionality*. It speeds up algorithms and makes visualization of high-dimensional data possible in 2D or 3D plots.